



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous Institute Affiliated to University of Mumbai)

Department of Computer Engineering
Generative AI Lab Manual AY 2026-27

Class: BTech(CE)
Course Name: Generative AI (PE III)
Faculty Name: Prof. Yogita Patil

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EXPERIMENT 7: Domain-Specific Generative AI Chatbot using a Pre-trained Language Model and Streamlit

1. Aim

To design and implement a domain-specific Generative AI chatbot using a pre-trained language model, prompt engineering, domain-specific knowledge, and a Streamlit-based user interface.

The same domain used in the previous LoRA/Fine-Tuning experiment must be used in this experiment.

2. Objectives

After completing this experiment, students will be able to:

1. Understand the architecture of a simple LLM-powered chatbot.
2. Use a pre-trained language model for text generation.
3. Provide domain-specific information to an LLM through prompting.
4. Design an effective system prompt for a domain-specific chatbot.
5. Develop a chatbot interface using Streamlit.
6. Connect a Python application to an LLM inference service.
7. Evaluate chatbot responses using domain-specific questions.
8. Identify limitations such as hallucination and incomplete knowledge.
9. Understand the difference between Fine-Tuning, Prompting, and RAG.
10. Convert a language model into a simple user-facing GenAI application.

3. Important Instruction

Use the Same Domain as the Previous Experiment

Students must use the same domain used in the previous LoRA/Fine-Tuning experiment.

For example:

Previous Experiment Domain	Experiment 7 Chatbot
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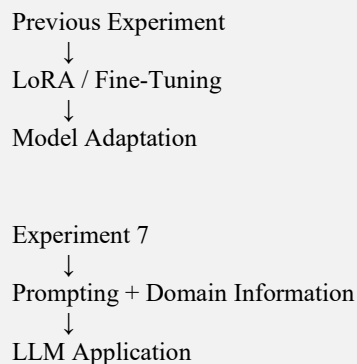
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Previous Experiment Domain	Experiment 7 Chatbot
HR Helpdesk	HR Helpdesk Chatbot
E-Commerce	E-Commerce Assistant
Healthcare	Healthcare Information Assistant
Banking	Banking Assistant
Education	Student Helpdesk
Travel	Travel Assistant

Students should not select a new domain.

The purpose is to compare two different approaches using the same domain:

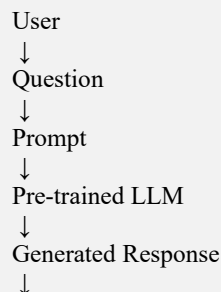


4. Conceptual Background

4.1 What is a Generative AI Chatbot?

A Generative AI chatbot uses a Large Language Model (LLM) to understand a user's question and generate a natural-language response.

Basic architecture:





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User

Unlike a traditional rule-based chatbot, an LLM-based chatbot can generate responses dynamically.

5. Domain-Specific Chatbot

A general-purpose LLM can answer questions from many domains.

For example:

User:

What is machine learning?

LLM:

Machine learning is a branch of AI...

However, organizations often need specialized assistants.

Examples:

HR Assistant
Banking Assistant
E-Commerce Assistant
College Assistant
Healthcare Assistant
Travel Assistant

The chatbot should be instructed to behave as an assistant for a particular domain.

6. Providing Domain Knowledge through Prompting

In this experiment, students will not fine-tune the model again.

Instead, relevant domain information will be supplied as part of the prompt.

For example:

You are an HR Helpdesk Assistant.

Domain Information:

Employees receive 12 casual leaves per year.
Leave requests should be submitted through the employee portal.
Work from home requires manager approval.
Salary is credited on the last working day of the month.

Answer the user's question using the information above.



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User:
How many casual leaves are available?

The LLM uses the information supplied in the prompt to generate the response.

7. Experiment 7 vs Previous LoRA Experiment

Students must understand the difference.

LoRA / Fine-Tuning	Experiment 7
Model is trained/adapted	Model is not trained
Model parameters are adapted	Model parameters remain unchanged
Training dataset required	Domain information supplied in prompt
Training process	Inference process
More computationally intensive	Lightweight
Model learns patterns from training	Model follows instructions/context
Produces an adapted model	Produces an application

8. Experiment 7 vs Experiment 8 – RAG

This distinction is particularly important.

Experiment 7	Experiment 8
Prompt-based chatbot	RAG system
No retriever	Retriever
No embeddings required	Embeddings used
No vector database required	Retrieval mechanism used
Domain information supplied directly	Relevant documents retrieved
Focus on application development	Focus on retrieval + generation
Streamlit interface	Retriever + Generator
Qualitative/basic evaluation	QA dataset + performance evaluation

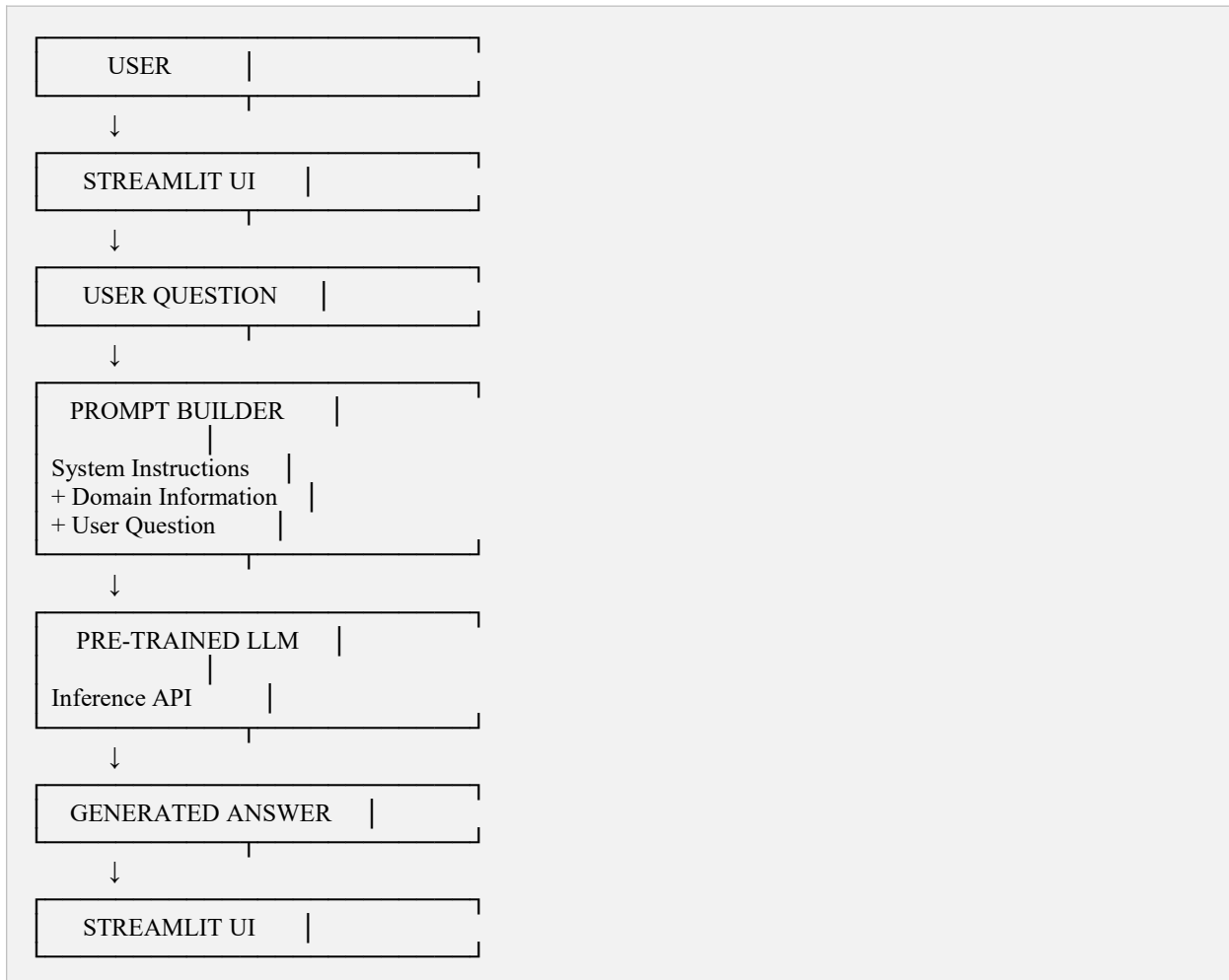
Therefore:



Experiment 7 is NOT a RAG experiment.

9. System Architecture

The system developed in this experiment will have the following architecture:



10. Software Requirements

- Google Colab
- Python 3.x
- Streamlit
- Hugging Face account
- Hugging Face inference access
- GitHub account



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- Web browser

No local GPU is required for the basic implementation.

11. Hardware Requirements

Students can perform the experiment using:

- Laptop/Desktop
- Internet connection
- Google account

The model inference is performed through an inference service rather than requiring students to train a model locally.

12. Step 1 – Open Google Colab

Open Google Colab and create a new notebook.

Rename the notebook:

```
EXP7_Domain_Specific_GenAI_Chatbot
```

13. Step 2 – Install Required Libraries

Run:

```
!pip install -q streamlit  
!pip install -q huggingface_hub
```

14. Step 3 – Select the Domain

Use the same domain used in the previous LoRA/Fine-Tuning experiment.

For example, if your previous experiment used:

```
HR Helpdesk
```

then your current experiment should also use:

```
HR Helpdesk
```



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15. Step 4 – Create Domain Information

Create a Python variable containing information about the selected domain.

Example – HR Domain

```
domain_information = """
HR HELP DESK INFORMATION

1. Employees are entitled to 12 casual leaves per year.

2. Employees should apply for leave through the employee portal.

3. Casual leave should normally be requested before the planned absence.

4. Work from home requires approval from the reporting manager.

5. Employees must complete the annual performance review process.

6. Salary is credited on the last working day of each month.

7. Employees should update their emergency contact information
through the employee portal.

8. Employees can contact HR for questions regarding employee benefits.

9. Medical leave may require supporting documentation.
"""
```

Important

Students must replace the above information with information related to their own previous experiment domain.

16. Step 5 – Obtain Hugging Face Access

Create/sign in to a Hugging Face account and obtain an access token for the available inference service.

Free-tier availability, model support, and usage limits may change, so students should use an instructor-approved currently available model/provider.

Never put the token directly into the source code.

Use:

```
from getpass import getpass
```



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```
HF_TOKEN = getpass("Enter your Hugging Face token: ")
```

The token will be hidden while entering it.

17. Step 6 – Connect Python to the LLM

Import the inference client:

```
from huggingface_hub import InferenceClient
```

Create the client:

```
client = InferenceClient(  
    token=HF_TOKEN  
)
```

Select an instructor-approved instruction/chat model currently available through the student's inference provider.

For example:

```
MODEL_NAME = "HuggingFaceH4/zephyr-7b-beta"
```

If the selected model is unavailable, use another currently supported instruction-following model.

18. Step 7 – Design the System Prompt

Create a prompt template.

```
def create_prompt(question):  
  
    prompt = f"""  
    You are a helpful domain-specific AI assistant.  
  
    Your domain is the selected domain from the previous experiment.  
  
    Use the following domain information when answering the question.  
  
    DOMAIN INFORMATION:  
    {domain_information}  
  
    INSTRUCTIONS:  
    1. Answer clearly and concisely.
```



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2. Use the domain information when relevant.
3. Do not intentionally invent domain-specific facts.
4. If the information is not available, clearly state that the information is not available in the provided information.
5. Do not answer questions outside the domain unless general information is explicitly requested.

USER QUESTION:
{question}

ANSWER:
""

return prompt

19. Step 8 – Generate the Response

Create a function:

```
def generate_answer(question):  
  
    prompt = create_prompt(question)  
  
    response = client.text_generation(  
        prompt,  
        model=MODEL_NAME,  
        max_new_tokens=200  
    )  
  
    return response
```

20. Step 9 – Test the Chatbot in Colab

Test the chatbot before creating the interface.

```
question = "How many casual leaves are available?"  
  
answer = generate_answer(question)  
  
print(answer)
```

Expected output:

Employees are entitled to 12 casual leaves per year.



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The exact response may vary depending on the model.

21. Step 10 – Test Multiple Questions

Students should test at least five questions.

For an HR chatbot:

Question 1

How many casual leaves are available?

Question 2

How should I apply for leave?

Question 3

Who needs to approve work from home?

Question 4

When is salary credited?

Question 5

How can I update my emergency contact?

The questions must be changed according to the student's selected domain.

22. Step 11 – Test an Unknown Question

Students must also ask a question that is not covered by the supplied domain information.

Example:

What is the capital of France?

or:

Who is the CEO of Microsoft?



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The chatbot should ideally indicate that this information is outside its supplied domain information.

Why perform this test?

It demonstrates an important limitation:

A prompt-based chatbot does not automatically become a reliable domain expert simply because domain information is included in the prompt.

23. Step 12 – Create the Streamlit Application

Create a Python file:

app.py

The basic application is:

```
import streamlit as st
from huggingface_hub import InferenceClient

st.set_page_config(
    page_title="Domain AI Assistant",
    page_icon="🤖"
)

st.title("🤖 Domain-Specific AI Assistant")

domain_information = """
Paste your domain-specific information here.
"""

HF_TOKEN = st.secrets["HF_TOKEN"]

MODEL_NAME = "YOUR_APPROVED_MODEL"

client = InferenceClient(
    token=HF_TOKEN
)

def create_prompt(question):

    return f"""
You are a helpful domain-specific AI assistant.

Use the following information to answer the question.

DOMAIN INFORMATION:
```



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```
{domain_information}
```

If the information is not available,
clearly say that it is not available.

USER QUESTION:
{question}

ANSWER:
""

```
def generate_answer(question):
```

```
    prompt = create_prompt(question)
```

```
    response = client.text_generation(  
        prompt,  
        model=MODEL_NAME,  
        max_new_tokens=200  
    )
```

```
    return response
```

```
question = st.text_input(  
    "Enter your question:"  
)
```

```
if st.button("Ask"):
```

```
    if question:
```

```
        answer = generate_answer(question)
```

```
        st.subheader("Assistant")
```

```
        st.write(answer)
```

```
    else:
```

```
        st.warning("Please enter a question.")
```

24. Step 13 – Run Streamlit

In Google Colab, save the application as:

```
app.py
```

Then run Streamlit.



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Students may use an approved tunneling method to expose the Streamlit application from the Colab runtime.

Example:

```
!streamlit run app.py &>/content/logs.txt &
```

The instructor can provide the preferred tunneling method for the current Colab environment.

25. Step 14 – Improve the Chatbot Interface

Students should modify the interface to include:

 Domain AI Assistant	
Domain: HR Helpdesk	
Ask your question:	
How many casual leaves...?	
[Ask Question]	
Assistant	
Employees are entitled to 12 casual leaves per year.	

Students may add:

- chatbot logo
- sidebar
- domain name
- conversation history
- clear-chat button
- example questions

26. Step 15 – Add Conversation History

Use Streamlit session state.

```
if "messages" not in st.session_state:
```



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```
st.session_state.messages = []
```

After generating an answer:

```
st.session_state.messages.append(  
    {  
        "question": question,  
        "answer": answer  
    }  
)
```

Display previous questions and answers:

```
for message in st.session_state.messages:  
  
    st.write("You:", message["question"])  
  
    st.write(  
        "Assistant:",  
        message["answer"]  
    )
```

27. Step 16 – Evaluate the Chatbot

Students must prepare at least 10 test questions.

Use different categories:

Category	Number
Direct domain questions	4
Paraphrased domain questions	2
Multi-part questions	2
Out-of-domain questions	2
Total	10

28. Evaluation Table

Students should complete:

Sr. No.	Question	Expected Answer Available?	Response Correct?	Hallucination/Unsupported Claim?
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Sr. No.	Question	Expected Answer Available?	Response Correct?	Hallucination/Unsupported Claim?
1	Domain question	Yes	Yes/No	Yes/No
2	Domain question	Yes	Yes/No	Yes/No
3	Domain question	Yes	Yes/No	Yes/No
4	Domain question	Yes	Yes/No	Yes/No
5	Paraphrased	Yes	Yes/No	Yes/No
6	Paraphrased	Yes	Yes/No	Yes/No
7	Multi-part	Yes	Yes/No	Yes/No
8	Multi-part	Yes	Yes/No	Yes/No
9	Out-of-domain	No	N/A	Yes/No
10	Out-of-domain	No	N/A	Yes/No

29. Required Experimentation

Students must perform at least two modifications.

Modification 1 – Prompt Engineering

Create two prompts.

Prompt A – Basic

Answer the following question:

{question}

Prompt B – Domain-specific

You are a domain-specific assistant.

Use the following domain information.

{domain_information}

Answer the question using the information provided.



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If the information is unavailable, clearly state that.

Question:
{question}

Compare the responses.

Modification 2 – Change the Domain Information

Add at least five additional domain-specific facts.

For example:

New Policy 1
New Policy 2
New Policy 3
New Policy 4
New Policy 5

Ask questions related to the new information.

Observe whether the model can use the newly supplied information.

Optional Modification 3 – Change the Model

If multiple approved models are available, students may compare two models using the same:

- prompt
- domain information
- questions

They should observe differences in:

- response quality
- response length
- relevance
- latency

30. Deployment – Streamlit Community Cloud

Students may deploy their chatbot using Streamlit Community Cloud.

The project should contain:

project/
|



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```
├── app.py
├── requirements.txt
└── README.md
```

requirements.txt

```
streamlit
huggingface_hub
```

If additional libraries are used, add them to requirements.txt.

31. API Key Security

Students must never upload the Hugging Face token to GitHub.

Incorrect

```
HF_TOKEN = "hf_XXXXXXXXXXXXXXXXXX"
```

Correct approach

```
HF_TOKEN = st.secrets["HF_TOKEN"]
```

The token should be stored using the application's secrets mechanism.

32. Expected Final Application

The final chatbot should allow a user to:

11. Enter a question.
12. Send the question to the LLM.
13. Generate a domain-specific response.
14. Display the response.
15. Continue asking questions.
16. Maintain conversation history.
17. Handle questions outside the supplied domain appropriately.

33. Sample Interaction

Example – HR Helpdesk

User:



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How many casual leaves are available?

Chatbot:

Employees are entitled to 12 casual leaves per year.

User:

How do I apply for leave?

Chatbot:

Employees should apply for leave through the employee portal.

User:

What is the capital of France?

Chatbot:

That information is outside the HR information provided to me.

34. Analysis Questions

Students must answer the following:

- Q1. What is a Generative AI chatbot?
- Q2. What is the role of the pre-trained LLM in this experiment?
- Q3. Why was the same domain used as in the previous experiment?
- Q4. How is this experiment different from LoRA fine-tuning?
- Q5. Does this experiment modify the parameters of the LLM?
- Q6. What is prompt engineering?
- Q7. Why is domain information included in the prompt?
- Q8. What happens when the user asks a question that is not present in the supplied information?
- Q9. What is hallucination?
- Q10. Can providing domain information in a prompt completely eliminate hallucination? Explain.
- Q11. What are the limitations of providing a large amount of domain information directly in a prompt?
- Q12. How would you improve this chatbot if the domain contained thousands of documents?
- Q13. What is the difference between this experiment and RAG?
- Q14. Why would RAG become useful when the knowledge base becomes very large?
- Q15. What is the role of Streamlit in this experiment?

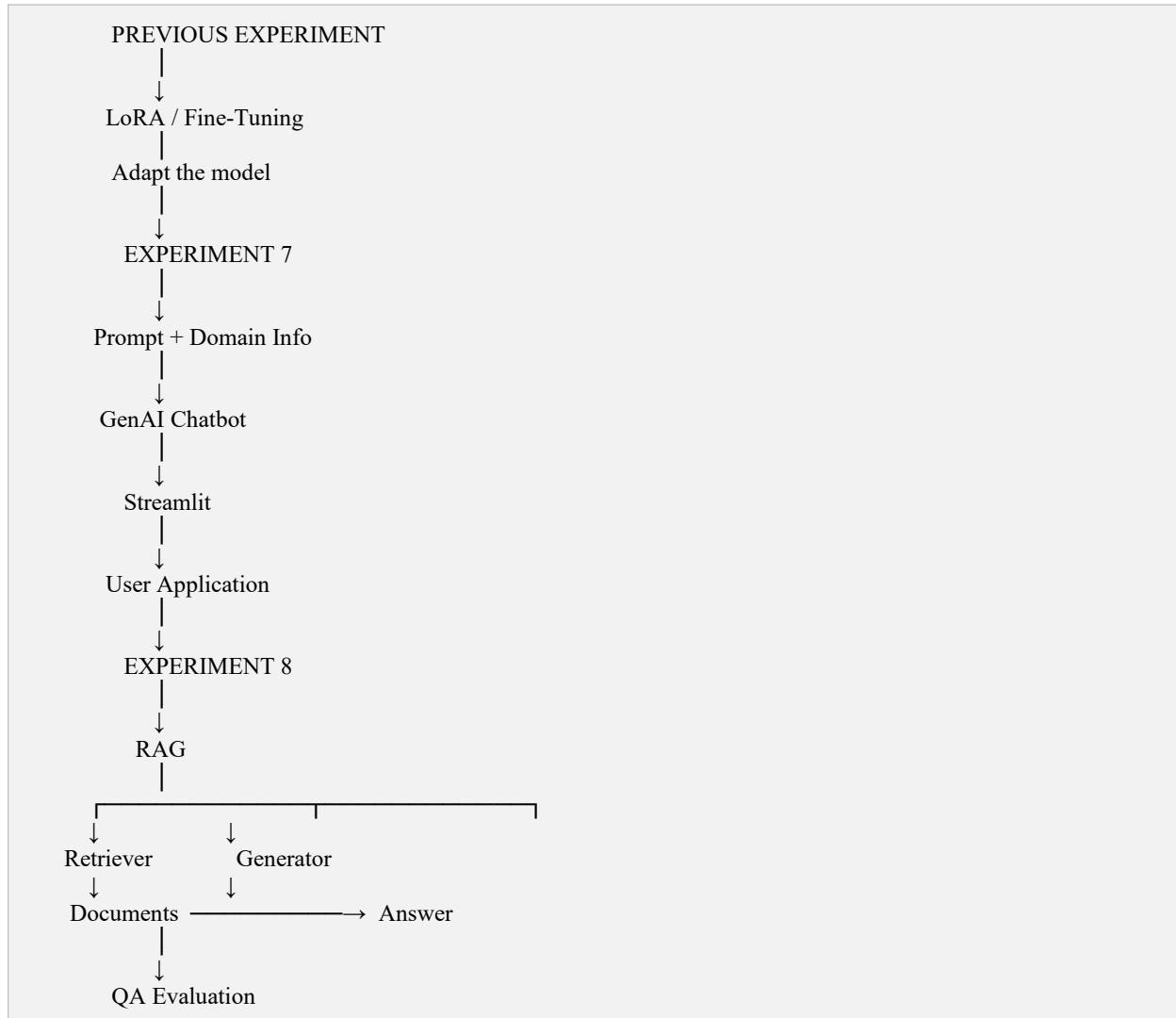
35. Comparison with Previous and Next Experiments

Students should be able to explain the following progression:



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36. Conclusion

In this experiment, a domain-specific Generative AI chatbot was developed using a pre-trained language model, prompt engineering, domain-specific information, and Streamlit.

The chatbot was built using the same domain used in the previous LoRA/Fine-Tuning experiment, allowing students to understand an alternative approach to building a domain-specific GenAI application without retraining the language model.

The experiment demonstrated how prompt design and domain-specific context can influence the responses generated by an LLM.

The chatbot was evaluated using domain-specific, paraphrased, multi-part, and out-of-domain questions.



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The experiment also highlighted important limitations of prompt-based systems, including hallucination, limited context size, and the difficulty of supplying large amounts of domain knowledge directly in every prompt.

These limitations motivate the use of Retrieval-Augmented Generation (RAG), which is introduced and evaluated separately in the next experiment.